



Module 1 — Computer Thinking

■ BEGINNER

- How computers process instructions
- Input → Process → Output model
- What is an algorithm?
- Flowcharts — visual logic
- Pseudocode — plan before coding

■ ADVANCED

- Big O notation basics
- Recursion vs iteration
- Dynamic programming concepts
- Algorithmic complexity analysis
- Design patterns overview

Module 2 — Python Basics

■ BEGINNER

- Variables & data types
- If / else statements
- Loops: for and while
- Functions & parameters
- Lists, dictionaries, tuples

■ ADVANCED

- List comprehensions
- Lambda functions
- Decorators & closures
- Generators & iterators
- Type hints & annotations

Module 3 — Python Projects

■ BEGINNER

- Reading & writing files
- Working with APIs (requests lib)
- Simple web scraping (BeautifulSoup)
- Basic data analysis (pandas)
- Automating tasks with Python

■ ADVANCED

- Building REST APIs (FastAPI/Flask)
- Database integration (SQLAlchemy)
- Async programming (asyncio)
- Testing with pytest
- Packaging & deployment

■ How to Write Your First Python Script

- 1 Install Python from python.org — choose version 3.10+
- 2 Open terminal/command prompt and type: `python --version` to verify
- 3 Create a new file: `hello.py` — open in VS Code or any text editor
- 4 Type: `print('Hello, World!')` and save the file
- 5 Run it: `python hello.py` — you should see Hello, World! printed

■ **Tip:** VS Code + Python extension is the best free setup for beginners — install it once and use it forever.